

WEB DEVELOPMENT (PHP + MySQL) — 60 ideas

1. **Student Management System** — Manage student profiles, courses, attendance and exam records with role-based access for admin/teachers/students. Tools: PHP, MySQL, Bootstrap, jQuery.
2. **Online Examination Portal** — Create timed exams with question banks, auto-grading MCQs and result reports. Tools: PHP, MySQL, JavaScript.
3. **Library Management System** — Track books, issue/return, fines, and searchable catalogue with admin and member panels. Tools: PHP, MySQL.
4. **Hostel / PG Finder System** — Listings, owner & tenant profiles, booking requests, and reviews for PG/hostels. Tools: PHP, MySQL, Google Maps API.
5. **Event Registration System** — Create events, manage registrations, tickets and attendee lists with PDF tickets. Tools: PHP, MySQL.
6. **Online Voting System** — Secure voting for college councils with authentication, one-vote enforcement and results dashboard. Tools: PHP, MySQL, sessions.
7. **College Feedback System** — Anonymized feedback collection, analysis and reporting for courses/teachers. Tools: PHP, MySQL, Chart.js.

8. **Alumni Management Portal** — Maintain alumni directory, event invites, donations and mentorship matching. Tools: PHP, MySQL.
9. **Job Portal for Freshers** — Employers post jobs/internships; candidates apply and upload resumes with filters. Tools: PHP, MySQL, file upload.
10. **Attendance Management System** — Record daily attendance, export CSV reports and summary analytics. Tools: PHP, MySQL, Bootstrap.
11. **Online Quiz Application** — Timed quizzes with categories, leaderboards and instant scoring. Tools: PHP, MySQL, AJAX.
12. **E-Commerce Website** — Product catalog, cart, checkout, payment integration and order management. Tools: PHP (Laravel optional), MySQL, Stripe/PayU.
13. **Online Grocery Store** — Category-based browsing, inventory, delivery slots and cart management. Tools: PHP, MySQL.
14. **Restaurant Ordering System** — Menu management, table orders, kitchen tickets and billing. Tools: PHP, MySQL, QR-code menus.
15. **Food Recipe Sharing Platform** — Users post recipes, rate, comment and save favorites. Tools: PHP, MySQL.

16. **Complaint Management System** — Log complaints, track status, assign to departments and report resolution stats. Tools: PHP, MySQL.
17. **Online Appointment Booking** — Book appointments with doctors/salons, calendar integration and email reminders. Tools: PHP, MySQL, FullCalendar.
18. **Hospital Management System** — Patient records, OP/IP management, billing and doctor schedules. Tools: PHP, MySQL.
19. **Pharmacy Inventory System** — Manage stock, expiry alerts and order re-stocking for pharmacies. Tools: PHP, MySQL.
20. **Online Course Registration** — Course catalog, enrollment, payment and certificate issuance. Tools: PHP, MySQL.
21. **Blood Donation Management System** — Donor registry, blood group search, donation drives and appointment scheduling. Tools: PHP, MySQL.
22. **NGO Donation Platform** — Campaign pages, donor management, donation receipts and impact reports. Tools: PHP, MySQL, payment gateway.
23. **Online Auction System** — Timed auctions with bidding, auto-extensions and bidder histories. Tools: PHP, MySQL, WebSockets (optional).

24. **Ticket Booking System** — Book event or travel tickets, seat selection and e-ticket generation. Tools: PHP, MySQL.
25. **Resume Builder Website** — Guided resume templates, export to PDF and profile sharing. Tools: PHP, MySQL, TCPDF.
26. **Blog Management System** — Create/edit posts, categories, comments and admin moderation. Tools: PHP, MySQL.
27. **Online Survey System** — Create surveys, collect responses and visualize results with charts. Tools: PHP, MySQL, Chart.js.
28. **Employee Payroll System** — Salary calculations, payslip generation and tax/deduction management. Tools: PHP, MySQL.
29. **Visitor Management System** — Record visitors, badges, check-in/out and visit logs. Tools: PHP, MySQL.
30. **Basic CRM System** — Track leads, customers, interactions and simple sales pipeline. Tools: PHP, MySQL.
31. **Online Insurance Portal** — Quote requests, policy listings, claim submission and status tracking. Tools: PHP, MySQL.
32. **Digital Notice Board** — Publish notices, schedules and alerts with date-based visibility. Tools: PHP, MySQL.

33. **Car Rental System** — Vehicle listings, booking calendar, pricing and admin fleet management. Tools: PHP, MySQL.
34. **Bike Rental System** — Short-term bookings, availability maps and rental history. Tools: PHP, MySQL.
35. **Movie Review Platform** — Submit reviews, ratings, and aggregate scorecards for films. Tools: PHP, MySQL.
36. **Travel Booking System** — Multi-city itineraries, bookings, and payment management for trips. Tools: PHP, MySQL.
37. **Online Hotel Reservation** — Room inventory, booking, cancellation and invoice generation. Tools: PHP, MySQL.
38. **Online Learning Portal** — Courses, video hosting (or links), quizzes and certificates. Tools: PHP, MySQL, video embed.
39. **Freelance Marketplace** — Project listings, proposals, escrow-like payment flow and messaging. Tools: PHP, MySQL.
40. **Classified Ads Website** — Post listings, search filters, contact via inbox and category moderation. Tools: PHP, MySQL.
41. **Lost & Found Portal** — Post lost/found items with images, contact and location tags. Tools: PHP, MySQL.

42. **Complaint Tracker for Society** — Residential society complaint logging, ticketing and maintenance allocations. Tools: PHP, MySQL.
43. **Online Result Publishing System** — Upload and display exam results with roll-number search and secure PDFs. Tools: PHP, MySQL.
44. **Online Portfolio Builder** — Templates to create personal portfolios and deployable portfolio pages. Tools: PHP, MySQL, HTML/CSS.
45. **Event Ticket Scanner System** — Admin issues tickets and a mobile web scanner verifies QR codes at entry. Tools: PHP, MySQL, QR library.
46. **Online Chat Application** — Real-time chat between users with rooms and message history. Tools: PHP with WebSockets or Node.js + Socket.io (frontend).
47. **Expense Tracker Website** — Track personal/club expenses, categories, and export monthly summaries. Tools: PHP, MySQL.
48. **Online File Management System** — Upload, organize, share files with role-based permissions. Tools: PHP, MySQL, filesystem or S3.
49. **Online Voting for College Council** — Secure polling module tailored to college elections with results audit. Tools: PHP, MySQL.

50. **Online Marketplace for Local Shops** — Onboard local shops, list products, local delivery and ratings. Tools: PHP, MySQL, maps API.
51. **Courier Tracking System** — Package status updates, tracking codes and route history for customers. Tools: PHP, MySQL.
52. **Online Examination Proctoring** — Basic proctoring: webcam snapshots, random question order and activity logs (privacy notice required). Tools: PHP, JavaScript.
53. **Online Banking Simulation** — Simulated accounts, transfers, statements and basic security practices for learning. Tools: PHP, MySQL.
54. **Real Estate Listing Website** — Property listings, search filters, agent contacts and image galleries. Tools: PHP, MySQL, maps API.
55. **Online Parking Slot Booking** — Reserve hourly slots, view occupancy map and manage payments. Tools: PHP, MySQL.
56. **College Placement Portal** — Companies post openings, students apply, schedule interviews and track placements. Tools: PHP, MySQL.
57. **Digital Attendance with QR** — Generate QR per class; students scan to mark attendance with time/location checks. Tools: PHP, MySQL, QR library.

58. **Online Feedback Analyzer** — Collect text feedback and show basic sentiment summary (keyword frequency). Tools: PHP, MySQL, simple NLP (Python microservice optional).
 59. **Medical Appointment Scheduler** — Doctor profiles, availability slots, reminders and basic teleconsult links. Tools: PHP, MySQL.
 60. **Online Internship Portal** — Companies post internships, students apply, resume upload and status tracking. Tools: PHP, MySQL.
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MACHINE LEARNING PROJECT IDEAS — 55 ideas

1. **House Price Prediction** — Regression model to predict house prices from location, size and features. Tools: Python, scikit-learn, pandas.
2. **Student Performance Prediction** — Predict student grades from attendance, assignments and past scores. Tools: Python, scikit-learn.
3. **Salary Prediction Model** — Estimate salary from skills, experience, education and location. Tools: Python, scikit-learn.
4. **Loan Approval Prediction** — Classifier to predict loan defaults or approvals using applicant data. Tools: Python, scikit-learn.

5. **Credit Card Fraud Detection** — Anomaly detection/classifier to flag suspicious transactions. Tools: Python, scikit-learn, imbalanced-learn.
6. **Fake News Detection** — Text classifier to flag misinformation using NLP features. Tools: Python, NLP (spaCy), scikit-learn.
7. **Email Spam Classifier** — Classify emails as spam/ham using text features and TF-IDF. Tools: Python, scikit-learn.
8. **Customer Churn Prediction** — Predict which customers will leave using historical behavior. Tools: Python, scikit-learn.
9. **Movie Recommendation System** — Collaborative/filtering model to recommend movies to users. Tools: Python, surprise or implicit.
10. **Music Recommendation System** — Recommend songs using user/item embeddings and collaborative filtering. Tools: Python, implicit.
11. **Product Recommendation Engine** — Item-to-item or user-based recommender for e-commerce suggestions. Tools: Python, scikit-learn.
12. **Stock Price Prediction** — Time-series forecasting with features like lagged returns (note: high risk). Tools: Python, statsmodels, sklearn.

13. **Sales Forecasting System** — Forecast product sales by store with time-series models. Tools: Python, Prophet or ARIMA.
14. **Weather Prediction Model** — Short-term forecasting using historical weather data. Tools: Python, time-series libs.
15. **Traffic Congestion Prediction** — Predict traffic levels from historical and event data. Tools: Python, scikit-learn.
16. **Disease Prediction System** — Predict disease likelihood from patient symptoms and history. Tools: Python, scikit-learn.
17. **Diabetes Prediction** — Classifier using medical attributes to predict diabetes risk. Tools: Python, scikit-learn.
18. **Heart Disease Prediction** — Predict heart disease from clinical measurements. Tools: Python, scikit-learn.
19. **Breast Cancer Detection** — Classifier for tumor benign/malignant using medical imaging features or tabular data. Tools: Python, scikit-learn.
20. **Customer Segmentation** — Cluster customers for targeted marketing using K-Means or hierarchical clustering. Tools: Python, scikit-learn.

21. **Face Recognition System** — Identify faces using embeddings and nearest-neighbor search (class project level). Tools: Python, OpenCV, face-recognition library.
22. **Handwritten Digit Recognition** — MNIST classification with classic ML or simple neural nets. Tools: Python, scikit-learn / TensorFlow.
23. **Text Sentiment Analysis** — Classify sentiment of reviews or tweets using TF-IDF and classifiers. Tools: Python, NLTK/spaCy, scikit-learn.
24. **Twitter Sentiment Analyzer** — Real-time tweet sentiment aggregation and trend tracking. Tools: Python, Tweepy, NLP.
25. **Resume Screening System** — Extract features from resumes to shortlist candidates automatically. Tools: Python, NLP, sklearn.
26. **Fake Review Detection** — Classify product reviews as genuine or fake using text patterns and metadata. Tools: Python, NLP.
27. **Chatbot using ML** — Intent classification and slot-filling chatbot for a domain (support/FAQ). Tools: Python, Rasa or simple sklearn models.
28. **Image Classification Model** — Classify images into categories using classical feature+ML or simple CNN. Tools: Python, TensorFlow/Keras.

29. **Speech Emotion Recognition** — Predict emotion from short audio clips using audio features. Tools: Python, librosa, sklearn.
30. **News Category Classifier** — Classify news articles into categories using text features. Tools: Python, scikit-learn.
31. **Online Exam Cheating Detection** — Use behavioral and clickstream features to detect suspicious activity. Tools: Python, anomaly detection.
32. **Movie Success Prediction** — Predict box-office success from cast, budget and social signals. Tools: Python, scikit-learn.
33. **YouTube Video Recommendation** — Suggest videos based on watch history and metadata. Tools: Python, collaborative filtering.
34. **Product Price Comparison ML** — Scrape prices, normalize, and recommend best deals. Tools: Python, scraping libs, simple models.
35. **Traffic Sign Recognition** — Classify traffic sign images for driver-assist projects. Tools: Python, CNN (Keras).
36. **Online Shopping Behavior Analysis** — Segment shoppers and predict next purchase using event logs. Tools: Python, pandas, scikit-learn.

37. **Employee Attrition Prediction** — Predict which employees may leave using HR features. Tools: Python, scikit-learn.
38. **Student Dropout Prediction** — Identify students at risk of dropping out using academic and engagement features. Tools: Python, scikit-learn.
39. **Malware Detection** — Classify binaries or network patterns as benign/malicious using extracted features. Tools: Python, static/dynamic analysis features, sklearn.
40. **SMS Spam Detection** — Classify SMS as spam/ham using text processing and classifiers. Tools: Python, scikit-learn.
41. **Face Mask Detection** — Classifier to check if person wears a mask using image data (classification). Tools: Python, OpenCV, Keras.
42. **Hand Gesture Recognition** — Recognize sign gestures from images or webcam frames for controls. Tools: Python, OpenCV, Keras.
43. **Loan Defaulter Prediction** — Predict loan default risk for banking decisions. Tools: Python, scikit-learn.
44. **Wine Quality Prediction** — Regression or classification to predict wine quality from physicochemical tests. Tools: Python, scikit-learn.

45. **Crop Yield Prediction** — Predict crop yields from features like weather, soil and inputs. Tools: Python, scikit-learn.
46. **Rainfall Prediction** — Time-series model for rainfall forecasting in a region. Tools: Python, time-series libs.
47. **Air Quality Index Prediction** — Forecast AQI from pollutant sensors and weather data. Tools: Python, scikit-learn.
48. **Recommendation for Online Courses** — Suggest courses based on user profile and past activity. Tools: Python, collaborative filtering.
49. **Plagiarism Detection** — Compare documents for similarity with text hashing and ML thresholds. Tools: Python, NLP.
50. **Market Basket Analysis** — Apriori/association rules to find product affinities in transaction data. Tools: Python, mlxtend.
51. **Recommendation for Jobs** — Match candidates to jobs using resume parsing and similarity scoring. Tools: Python, NLP.
52. **Online Fraud Detection** — Detect fraudulent behavior patterns in e-commerce or payments. Tools: Python, anomaly detection.

53. **Predictive Maintenance** — Predict equipment failure using sensor time-series and classification. Tools: Python, time-series/ML.
 54. **Social Media Trend Analysis** — Detect emerging topics and trend shifts from posts. Tools: Python, NLP, topic modeling.
 55. **Music Genre Classification** — Classify songs into genres from audio features. Tools: Python, librosa, sklearn/Keras.
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DESKTOP SOFTWARE PROJECT IDEAS — 50 ideas

1. **College Management Desktop App** — Full offline ERP for college: students, staff, fees and exams. Tools: C#/Java, SQLite.
2. **Student Result Analysis Tool** — Input marks, generate statistics, graphs and topper lists. Tools: Python (Tkinter) or C#, matplotlib.
3. **Desktop Attendance Manager** — Record attendance, export reports and print registers offline. Tools: Python, SQLite.
4. **Offline Exam Software** — Proctored offline exam app with timed tests and local result storage. Tools: Java/Python, local DB.

5. **Desktop Library Automation** — Manage books, borrowers and due reminders on a desktop. Tools: Java, SQLite.
6. **Smart Timetable Generator** — Generate conflict-free timetables based on constraints and preferences. Tools: Python, constraint-solver.
7. **Faculty Scheduling Software** — Assign faculty to classes, handle substitutions and clash detection. Tools: C#/Java.
8. **Hostel Management Desktop App** — Room allocation, maintenance requests and billing offline. Tools: Python, SQLite.
9. **Desktop Inventory Manager** — Stock tracking, reorder alerts and invoice printing for small stores. Tools: C#, SQL.
10. **Payroll Calculation Software** — Salary computation with tax rules and payslip generation. Tools: C#/Java, CSV export.
11. **Invoice Generator Software** — Create, save and print invoices with templates and customer history. Tools: Python, reportlab.
12. **Offline Billing System** — POS-like billing for shops with barcode support and day-end reports. Tools: Java/C#.

13. **Desktop CRM Tool** — Local CRM for small businesses to track customers and interactions. Tools: C#, SQLite.
14. **Student Progress Tracker** — Visualize student progress across semesters and subjects. Tools: Python, matplotlib.
15. **Desktop Resume Builder** — Create/resume templates offline and export to PDF. Tools: Python, reportlab.
16. **File Encryption Tool** — Encrypt/decrypt files with password protection and simple UI. Tools: Python, cryptography.
17. **Password Vault Software** — Store credentials encrypted locally with a master password. Tools: Python, cryptography.
18. **Desktop Backup Tool** — Schedule and run folder backups to local/network drives. Tools: Python, OS libs.
19. **Desktop Chat Application** — LAN-based chat client-server for internal networks. Tools: Java/C#, sockets.
20. **Desktop Task Automation Tool** — Create rules to automate file moves, renames or periodic tasks. Tools: Python.
21. **Offline Quiz Application** — Host quizzes for classes with local result scoring and reports. Tools: Python, SQLite.

22. **Desktop Expense Tracker** — Track daily expenses, budgets and monthly summaries offline. Tools: Python, SQLite.
23. **Desktop Reminder System** — Scheduling reminders, alarms and to-do lists with notifications. Tools: Python.
24. **Desktop Alarm & Timer** — Multiple alarms/timers with custom sounds and snooze. Tools: Python.
25. **Desktop Email Automation Tool** — Schedule and send bulk templated emails via SMTP with logs. Tools: Python, smtplib.
26. **Desktop File Organizer** — Automatically sort files into folders based on rules/metadata. Tools: Python.
27. **Desktop Media Player** — Play audio/video with playlist, equalizer and basic metadata. Tools: C#/Electron.
28. **Desktop PDF Utility Tool** — Merge, split, compress PDFs and extract pages. Tools: Python, PyPDF2.
29. **Desktop Attendance using Face** — Face-recognition-based local attendance recorder using webcam. Tools: Python, OpenCV.
30. **Desktop Weather App** — Weather widget that fetches (or simulates) forecasts and stores favorites. Tools: Electron/Python.

31. **Desktop Calendar Planner** — Events, reminders and printable monthly planners. Tools: C#/Java.
32. **Desktop Chatbot Assistant** — Local assistant for notes, simple Q&A and task scheduling. Tools: Python, NLP libs.
33. **Desktop Exam Result Publisher** — Generate and print formatted result sheets and certificates. Tools: Python.
34. **Desktop Code Editor** — Lightweight code editor with syntax highlighting and run support. Tools: Electron, Ace editor.
35. **Desktop Compiler Tool** — Wrapper to compile/run code in multiple languages locally. Tools: Electron, child processes.
36. **Desktop Student Analytics Tool** — Analyze student datasets and visualize key metrics offline. Tools: Python, pandas.
37. **Desktop Notification Manager** — Centralized app to manage system notifications and schedules. Tools: Electron.
38. **Desktop Data Visualization Tool** — Load CSVs and create charts/plots interactively. Tools: Python, matplotlib.

39. **Desktop Search Engine** — Index local files and provide fast search with previews. Tools: Python, Whoosh.
40. **Desktop Encryption Utility** — Encrypt folders, create secure archives and password-protect them. Tools: Python, cryptography.
41. **Desktop File Sync Tool** — Sync selected folders between two local locations or drives. Tools: Python, rsync wrapper.
42. **Desktop Chat Server** — Host a simple chat server and manage connected clients locally. Tools: Node.js or Python (asyncio).
43. **Desktop Data Entry Tool** — Forms and validation for repetitive data entry tasks with export. Tools: C#/Python.
44. **Desktop Report Generator** — Create templated reports from data inputs and export PDFs. Tools: Python, reportlab.
45. **Desktop School ERP** — Modular ERP for small schools: attendance, fees, exams and notices. Tools: Java/C#.
46. **Desktop Feedback Analyzer** — Import feedback text and run local sentiment/keyword analysis. Tools: Python, NLTK.

47. **Desktop Survey Tool** — Build surveys, collect responses and analyze results locally. Tools: Python.
 48. **Desktop Helpdesk Tool** — Ticket creation, assignment and status tracking for small IT teams. Tools: C#/Python.
 49. **Desktop Log Analyzer** — Parse and visualize server/log files to find anomalies. Tools: Python.
 50. **Desktop Monitoring System** — Monitor system metrics (CPU, disk, network) and alert on thresholds. Tools: Python.
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DEEP LEARNING PROJECT IDEAS — 50 updated ideas

1. **Smart Attendance using Face Recognition** — Use face embeddings to mark attendance automatically from webcam images. Tools: Python, OpenCV, FaceNet/ArcFace, Keras/PyTorch.
2. **Face Mask Detection System** — Detect mask/no-mask on faces in images or live video for compliance monitoring. Tools: Python, Keras/TensorFlow, OpenCV.
3. **Student Emotion Detection from Facial Images** — Classify emotional states (happy, sad, neutral) from student webcam frames to gauge engagement. Tools: Python, CNNs, Keras.

4. **Handwritten Answer Sheet Digit Recognition** — OCR and digit recognition for individual handwritten answers to assist grading. Tools: Python, CNNs, Tesseract (hybrid).
5. **Automatic Answer Sheet Marks Extraction** — Detect and extract marks/answers from scanned answer sheets using OCR + layout parsing. Tools: Python, OCR, OpenCV.
6. **License Plate Recognition (Indian Vehicles)** — Detect plates and read characters considering regional formats and fonts. Tools: Python, OpenCV, OCR, deep learning.
7. **Traffic Signal Violation Detection** — Detect vehicles violating signals using object detection and tracking from CCTV. Tools: Python, YOLO/Detectron, OpenCV.
8. **Crowd Density Estimation System** — Estimate people count & density in crowded scenes with regression-based CNNs. Tools: Python, Keras, crowd-count datasets.
9. **Smart CCTV Intrusion Detection** — Detect unusual movement or intrusion in restricted zones using activity recognition. Tools: Python, deep learning, OpenCV.
10. **Online Exam Face Authentication** — Verify examinee identity by matching live face to registered photo before exam start. Tools: Python, face embeddings.
11. **Driver Drowsiness Detection System** — Detect eye closure and yawns to warn drowsy drivers in real-time. Tools: Python, OpenCV, CNN/RNN.

12. **Voice-Based User Authentication** — Speaker recognition system using voice embeddings for secure authentication. Tools: Python, librosa, deep speaker models.
13. **Speech-to-Text Note Converter** — Convert lecture audio into searchable transcripts with punctuation. Tools: Python, pretrained ASR models.
14. **Automatic MCQ Question Generator** — Generate multiple-choice questions from text using NLP and transformer models. Tools: Python, Hugging Face Transformers.
15. **Essay Grading System using Deep Learning** — Score essays on content and language using sequence models and embeddings. Tools: Python, transformers.
16. **Resume Skill Extraction System** — Extract skills and experience from resumes using named-entity recognition (NER). Tools: Python, spaCy, transformers.
17. **College Enquiry Chatbot (Rule + DL Based)** — Hybrid chatbot answering college FAQs with intent classification and retrieval. Tools: Python, Rasa/transformers.
18. **Image-Based Plant Disease Detection** — Identify plant diseases from leaf images and suggest treatments. Tools: Python, CNNs, Keras.

19. **Crop Growth Stage Detection using Images** — Classify crop growth stages from field images to inform agronomy decisions. Tools: Python, CNNs.
20. **Medical Image Classification System** — Classify X-rays/CT/MRI images to detect specific conditions. Tools: Python, Keras/PyTorch, transfer learning.
21. **Chest X-Ray Disease Detection** — Multi-label classifier for conditions like pneumonia or TB from chest X-rays. Tools: Python, deep CNNs, transfer learning.
22. **Brain Tumor Detection using MRI** — Segment or classify brain tumor presence using MRI slices and U-Net architectures. Tools: Python, Keras, medical datasets.
23. **Skin Disease Classification System** — Classify dermatological images into disease categories using CNNs. Tools: Python, transfer learning.
24. **Emotion Detection from Voice Signals** — Classify speaker emotions from audio using spectrograms + CNN/RNN. Tools: Python, librosa, Keras.
25. **Fake Voice / Spoof Detection System** — Detect synthesized or replayed audio to prevent voice spoofing. Tools: Python, deep learning for audio.
26. **Sign Language Recognition System** — Translate hand signs from video to text using CNNs+LSTMs or transformers. Tools: Python, OpenCV, deep models.

27. **Hand Gesture Recognition System** — Recognize gestures for UI control or assistive tech using CNNs.
Tools: Python, OpenCV, Keras.
28. **Face Anti-Spoofing Detection** — Distinguish real faces from photos or videos with texture and motion features. Tools: Python, CNNs.
29. **Video Surveillance Activity Recognition** — Classify complex activities in video (fighting, falling, loitering).
Tools: Python, 3D CNNs or I3D.
30. **Road Accident Detection from CCTV** — Detect accidents automatically in traffic video using detection + anomaly logic. Tools: Python, object detection, tracking.
31. **Vehicle Counting and Classification** — Count vehicles and classify types (car, bus, truck) from road cameras. Tools: Python, YOLO/SSD.
32. **OCR for Handwritten Notes** — Convert handwritten classroom notes to editable text using sequence models. Tools: Python, OCR + deep models.
33. **Printed Text OCR System** — High-accuracy OCR pipeline for printed documents with layout analysis.
Tools: Python, Tesseract + deep correction.
34. **Document Classification System** — Classify documents (invoices, letters) and auto-route to departments. Tools: Python, transformers.

35. **News Image + Text Categorization** — Jointly classify news items using both headline text and image. Tools: Python, multimodal transformers.
36. **Image Caption Generator** — Produce natural-language captions for images using encoder-decoder CNN+RNN or transformers. Tools: Python, Keras, transformers.
37. **Deepfake Image Detection System** — Detect manipulated images/videos using forensic and deep features. Tools: Python, CNNs.
38. **Product Recommendation System using Neural Networks** — Use embeddings and neural architectures for personalized product suggestions. Tools: Python, Keras.
39. **Student Performance Prediction using Deep Learning** — Use neural nets on temporal academic data to forecast performance. Tools: Python, LSTM/transformers.
40. **Exam Question Difficulty Prediction** — Predict difficulty level of questions using embeddings and past performance data. Tools: Python, NLP models.
41. **Online Review Sentiment Scoring** — Fine-grained sentiment scoring with transformer-based models. Tools: Python, Hugging Face.

42. **Chatbot Intent Classification System** — Robust intent classifier using transfer learning for complex chatbots. Tools: Python, transformers.
43. **Human Activity Recognition from Video** — Detect activities like walking, running, falling using video models. Tools: Python, 3D CNNs.
44. **Fall Detection System for Elderly** — Detect falls in real-time and trigger alerts using video or wearable sensors. Tools: Python, deep models, edge deployment.
45. **Face Recognition Door Lock System** — Edge-deployable face recognition controlling physical door locks (simulation). Tools: Python, OpenCV, embeddings.
46. **Real-Time Object Detection System** — Live detection with bounding boxes for multiple classes (YOLO/SSD). Tools: Python, YOLOv5/YOLOv8.
47. **Emotion-Aware Recommendation System** — Use detected user emotion to personalize content recommendations. Tools: Python, multimodal models.
48. **Voice-Controlled Desktop Assistant** — Speech recognition + intent models to control desktop tasks. Tools: Python, speech models.
49. **Video Content Classification System** — Tag videos by content categories for moderation or search. Tools: Python, CNN+RNN or transformers.

50. **Image Similarity Search Engine** — Retrieve visually similar images using embeddings and nearest-neighbor indexing. Tools: Python, FAISS, CNN embeddings.